

Roadmap: Complete Machine Learning & Data Science Program

Phase 1: Programming Foundation

Goal: Build Python proficiency, understand programming logic & file handling.

- Python Basics (Syntax, Data Types, Loops, Functions, OOPs)
- Working with Files (Text/JSON)
- Inventory System Projects
- Jupyter Notebook Setup
- Learn Standard Libraries: OS, NumPy basics

Phase 2: Data Handling & Visualization

Goal: Learn to manipulate and visualize data effectively.

- Master NumPy & Pandas
- Data Preprocessing
- Exploratory Data Analysis (EDA)
- Matplotlib & Seaborn for Visualization
- Streamlit for Web Dashboards

Phase 3: Mathematics & Statistics for ML

Goal: Understand the mathematical core behind algorithms.

- Linear Algebra (Vectors, Matrices, Eigenvalues)
- Vector Norms & Distance Metrics
- Calculus (Derivatives, Gradients)
- Optimization (Gradient Descent)
- Statistics (Descriptive, Inferential, Sampling, CLT)
- Bias-Variance, Outlier Detection

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Phase 4: Data Analytics & Tools

Goal: Analyze data using business tools & SQL.

- Excel (Pivot Tables, Cleaning, Analysis)
- Power BI Dashboards
- SQL (DML, Joins, Subqueries, Window Functions)
- Web Scraping with Selenium
- Data Cleaning Projects

Phase 5: Machine Learning

Goal: Learn & implement core ML algorithms.

- Regression (Linear, Polynomial, Multiple)
- Classification (SVM, Decision Trees, Random Forests)
- Clustering Algorithms (K-Means, etc.)
- Feature Engineering Techniques
- Evaluation Metrics

Phase 6: AI Specialization

Goal: Dive deeper into advanced AI technologies.

- Deep Learning: Perceptrons, MLP, CNN
- Image Processing: OpenCV, Webcam Capture, Filters
- NLP: Spacy, Regex, Word Clouds, Text Analysis
- Projects like Face Detection, Text Summarization

Phase 7: Generative AI

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Goal: Explore cutting-edge GenAI tools & use-cases.

- What is GenAI? Prompt Engineering
- OpenAI / Hugging Face API Usage
- Text Generation (LLMs)
- Image Generation (DALL·E, Stable Diffusion)
- LangChain Chatbot Building
- Ethics & Responsible Use

Phase 8: Projects & Deployment

Goal: Apply knowledge through real-world projects.

- Data Projects: Sales Analysis, COVID Dashboard, Web Scrapers
- ML Projects: House Prices, Diabetes Prediction, Customer Segmentation
- NLP/Image Projects: Selfies Capture, Spell Checker, Instagram Filters
- Streamlit/Power BI Dashboards

Tips for Progress

- Revise math/stats regularly
- Build mini-projects for each phase
- Maintain a portfolio (GitHub, Kaggle)
- Participate in mock interviews
- Attend mentorship/live sessions for extra projects